

Injective and Surjective Functions

Definition. A function $f : A \rightarrow B$ is:

1. **injective** (or *one-to-one*) if for all $a, a' \in A$, $f(a) = f(a') \implies a = a'$; $(\forall a, a' \in A, f(a) = f(a') \implies a = a')$
2. **surjective** (or *onto* B) if for every $b \in B$ there is an $a \in A$ with $f(a) = b$; $(\forall b \in B \exists a \in A, b = f(a))$
3. **bijective** if f is both injective and surjective.

Note: f is not one-to-one $\iff \exists a, a' \in A, f(a) = f(a')$, but $a \neq a'$
 f is not onto $\iff \exists b \in B \forall a \in A, b \neq f(a)$

01. How to show a function $f : A \rightarrow B$ is injective:

Direct approach:

Suppose $a, a' \in A$ and $f(a) = f(a')$.

$\vdots$

Therefore $a = a'$.

Contrapositive approach:

Suppose $a, a' \in A$ and $a \neq a'$.

$\vdots$

Therefore $f(a) \neq f(a')$.

02. How to show a function $f : A \rightarrow B$ is surjective:

Suppose $b \in B$.

[Prove there exists $a \in A$ for which $f(a) = b$.]

Example 01.

Prove that $f : [0, +\infty[\rightarrow \mathbb{R}$ defined by $f(x) = x^2$ is injective.

Proof: Let $x_1, x_2 \in [0, +\infty[$.

Suppose that $f(x_1) = f(x_2)$.

Then, $x_1^2 = x_2^2$.

Therefore, $|x_1| = |x_2|$

$x_1 = x_2$ (Since, $x_1, x_2 \geq 0$)

Thus, f is injective. ■

Example 02.

Prove that $f : \mathbb{R} \rightarrow [0, +\infty[$ defined by $f(x) = x^2$ is surjective.

Proof: Let $y \in [0, +\infty[$.

Choose $x = \sqrt{y}$.

Then, $x \in \mathbb{R}$ and $f(x) = (\sqrt{y})^2 = y$.

Hence, $\forall y \in [0, +\infty[, \exists x = \sqrt{y} \in \mathbb{R}, y = f(x)$.

Therefore, f is surjective. ■

Example 03.

Prove that $f : [0, +\infty[\rightarrow [0, +\infty[$ defined by $f(x) = x^2$ is bijective.

Example 04.

Show that the function $f : \mathbb{R} - \{0\} \rightarrow \mathbb{R}$ defined as $f(x) = \frac{1}{x} + 1$ is injective but not surjective.

Example 05.

Show that the function $f : \mathbb{R} - \{0\} \rightarrow \mathbb{R} - \{1\}$ where $f(x) = \frac{1}{x} + 1$ is injective and surjective (hence bijective).

Theorem Suppose $f : A \rightarrow B$ and $g : B \rightarrow C$. As we saw, it follows that $g \circ f : A \rightarrow C$.

1. If f and g are both one-to-one, then so is $g \circ f$.
2. If f and g are both onto, then so is $g \circ f$.

Proof

Exercises

01. Let $A = \mathbb{R} \setminus \{1\}$, and let $f : A \rightarrow A$ be defined as follows:

$$f(x) = \frac{x+1}{x-1}.$$

- (a) Show that f is one-to-one and onto.
 - (b) Show that $f \circ f = i_A$.
02. Suppose $f : A \rightarrow B$ and $g : B \rightarrow C$.
- (a) Prove that if $g \circ f$ is onto then g is onto.
 - (b) Prove that if $g \circ f$ is one-to-one then f is one-to-one.
03. Suppose $f : A \rightarrow B$ and $g : B \rightarrow C$.
- (a) Prove that if f is onto and g is not one-to-one, then $g \circ f$ is not one-to-one.
 - (b) Prove that if f is not onto and g is one-to-one, then $g \circ f$ is not onto.
04. Suppose A , B , and C are sets and $f : A \rightarrow B$.
- (a) Prove that if f is onto, $g : B \rightarrow C$, $h : B \rightarrow C$, and $g \circ f = h \circ f$, then $g = h$.
 - (b) Suppose that C has at least two elements, and for all functions g and h from B to C , if $g \circ f = h \circ f$ then $g = h$. Prove that f is onto.
05. Suppose A , B , and C are sets and $f : B \rightarrow C$.
- (a) Prove that if f is one-to-one, $g : A \rightarrow B$, $h : A \rightarrow B$, and $f \circ g = f \circ h$, then $g = h$.
 - (b) Suppose that $A \neq \emptyset$, and for all functions g and h from A to B , if $f \circ g = f \circ h$ then $g = h$. Prove that f is one-to-one.

Definition

Let $\varphi : A \rightarrow B$ be a function and $C \subseteq A$.

- The **restriction** of φ to C is the function $\varphi|_C : C \rightarrow B$ so that

$$\varphi|_C(x) = \varphi(x) \text{ for all } x \in C.$$

- The function $\psi : D \rightarrow E$ is an **extension** of φ if $A \subseteq D$, $B \subseteq E$, and $\psi|_A = \varphi$.

Example 1: Polynomial Function

Let

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2.$$

Restriction

Restrict the domain to $[0, \infty)$.

$$g = f|_{[0, \infty)} : [0, \infty) \rightarrow \mathbb{R}, \quad g(x) = x^2.$$

Here, g is called the **restriction** of f to $[0, \infty)$.

Extension

Conversely, suppose we start with

$$g : [0, \infty) \rightarrow \mathbb{R}, \quad g(x) = x^2.$$

Then

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2$$

is an **extension** of g .

Example 2: Trigonometric Function

Let

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = \sin x.$$

Restriction

Restrict the domain to

$$[0, \pi].$$

Then

$$f|_{[0, \pi]} : [0, \pi] \rightarrow \mathbb{R}, \quad x \mapsto \sin x.$$

Extension

Starting with

$$g : [0, \pi] \rightarrow \mathbb{R}, \quad g(x) = \sin x,$$

an extension is

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = \sin x.$$

Example 3: Absolute Value Function

$$\text{Let } f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = |x|.$$

Restriction

Restrict the domain to the integers:

$$f|_{\mathbb{Z}} : \mathbb{Z} \rightarrow \mathbb{R}, \quad f(n) = |n|.$$

Extension

The function

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = |x|$$

is an extension of $f|_{\mathbb{Z}}$.

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